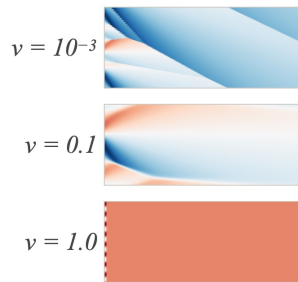


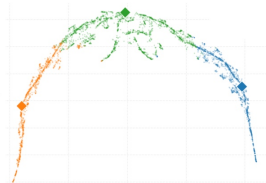
PHASE 1 · OFFLINE — REGIME DISCOVERY

1 Snapshots



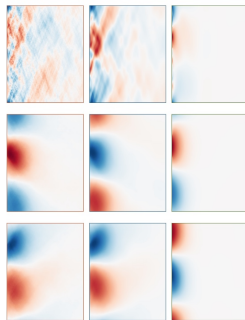
$$D = \{s^{(i)}, u_0^{(i)}, \kappa^{(i)}\}$$

2 EM regimes



soft regime labels γ_{im}

3 Per-regime bases

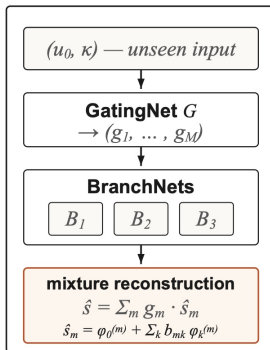


$$\{\Phi_m, \pi_m\}, m = 1 \dots M$$

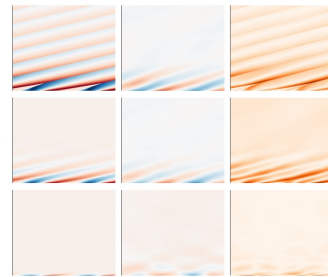
frozen regime bases
 $\{\Phi_m, \pi_m\} \rightarrow B_m$

PHASE 2 · ONLINE — GATED OPERATOR

4 Gated operator



5 Reconstruction



$\hat{s}(x, t)$ vs. truth · |error|